ClaimPilot AI — Architecture Overview

AI-assisted insurance claim verification. Customers file claims and upload PDFs directly to private Amazon S3; Amazon Bedrock reads each document, rules compare it with the claim, and a person makes every decision. Built without the AWS SDK on an AWS Free-plan account (ap-southeast-2).

⑬ Claim status & AI results (read path: Next.js → API Gateway → Lambda → DynamoDB)

1 Customer Portal

Next.js · React · TypeScript · Tailwind CSS

Next.js server

Route handlers + server components.
Holds the API key server-side only;
the browser never sees it.

✓ Implemented & tested · Runs locally

② HTTPS + API key

① HTTPS: claim data ⇄ presigned URL, claim

Customer browser

Claim form, policy details, PDF upload.
Client-side validation + PDF check.
No browser storage of files or data.

✓ Implemented & tested · Runs locally

④ Secure direct PDF upload (presigned PUT · 5 min · locked size + type)

Portal hosting

Runs locally (npm run dev).
Not deployed publicly.

○ Optional · not used

2 AWS API & Storage

IAM · API Gateway · Lambda · S3 · DynamoDB

API Gateway (REST)

HTTPS entry point · API key required.
Usage plan: 5 req/s, 500 req/day.
Requests without key → 403.

✓ Implemented & tested · Uses credits

③ Lambda proxy invoke

Lambda · claimpilot-api

Validates input · presigns uploads
(hand-written SigV4, no SDK) · stores
claims · strips internal AWS errors.

✓ Implemented & tested · Always Free tier

⑤ Verify upload (first 5 bytes)

Amazon S3

claim-pilot · private · SSE-S3.
HTTPS-only · CORS for the portal.
Uploads expire after 30 days.

✓ Implemented & tested · Uses credits

⑥ Claim metadata PutItem / GetItem / Scan

Amazon DynamoDB

claimpilots-claim · provisioned 5/5.
Claims, results, usage counters.
Conditional writes stop duplicates.

✓ Implemented & tested · Always Free tier

⑫ Processing result (UpdateItem, owner only)

⑨ Read PDF (GetObject)

AWS IAM

2 least-privilege Lambda roles:
one prefix, one table, two models.
No delete permissions.

✓ Implemented & tested · No charge

CloudWatch Logs

Lambda logs with request IDs and
token counts · 3-day retention.
AWS error details stay here.

✓ Implemented & tested · Always Free tier

3 AI Verification Workflow

DynamoDB Streams · Lambda · Amazon Bedrock

⑦ INSERT event

DynamoDB Streams

Emits an event for each new claim.
Filter: INSERT of CLM-* items only,
so results never re-trigger.

✓ Implemented & tested · Always Free tier

⑧ Trigger (new CLM-* only)

Step Functions · EventBridge · SQS · SNS

Not needed: Streams → Lambda gives
retries + filtering. Free eligibility
not verified, so not provisioned.

○ Optional · not used

Lambda · claimpilot-processor

Locks claim · monthly cap (50) · page
limit (3) · rule checks: policy, date,
amount · saves the assessment.

✓ Implemented & tested · Always Free tier

× denied

Amazon Textract

Planned for OCR. AccessDenied:
explicit deny in the Free plan's
service control policy (verified).

× Blocked on Free plan

⑩ PDF → fields + quotes

⑪ fields + rule results → summary

Bedrock · Nova Lite

Reads the PDF (Converse document).
Returns fields + exact quotes +
clear/unclear flag. No free tier.

✓ Implemented & tested · Uses credits

Bedrock · Nova Micro

Writes the reviewer summary and
answers: does the document fit the
incident? (yes / no / unclear)

✓ Implemented & tested · Uses credits

Safety rules

Rules decide matches, never the model.
Only outcomes: ready for standard
review / needs human review.

✓ Implemented & tested

4 Results Dashboard

Status · findings · human review

Customer dashboard

Totals, open, flagged for review.
Recent claims with status and
automated-check result (live).

✓ Implemented & tested · Runs locally

⑭ Open claim

Claim details

Extracted fields + quotes, missing
info, inconsistencies, AI summary,
review status, Try again on errors.

✓ Implemented & tested · Runs locally

⑮ Findings for human review

Human claims handler

Makes every decision and sets the
claim status (DynamoDB console;
no reviewer UI yet).

✋ Manual process

Arrows

- HTTPS request / API call
- Secure document upload
- Event trigger (stream)
- Processing result / status
- Blocked on this plan

Implementation status

- ✓ Implemented & tested
- ✋ Manual process
- × Blocked on Free plan
- ○ Optional · not used

Cost on the Free plan ($0 billed)

- Always Free tier — within monthly limits (Lambda, DynamoDB, Streams, Logs)
- Uses credits — no free tier: Bedrock, API Gateway, S3 (tiny, from approved credits)

Guards: 500 API requests/day · 50 AI checks/month · PDFs ≤ 3 pages · MONTHLY_LIMIT=0 switches AI off

Stages

1. 1 Customer Portal
2. 2 AWS API & Storage
3. 3 AI Verification Workflow
4. 4 Results Dashboard

Key guarantees

- PDFs never pass through the web server
- No AWS credentials or API key in the browser
- Repeated submissions return the same claim
- Each claim is processed exactly once
- AI never approves, rejects or alleges fraud

Verified on real AWS (2026-10-29): end-to-end tests 12/12 · unit tests 24/24. Details: README.md · docs/architecture.md